13 Show that $6 + \left[(x + 5) \div \dfrac{x^2 + 3x - 10}{x - 1} \right]$ simplifies to $\dfrac{ax - b}{cx - d}$ where a, b, c and d are integers.

$x^2 + 3x - 10 = (x + 5)(x - 2)$

1 mark
Factorise the quadratic.

$6 + (x + 5) \div [(x + 5)(x - 2)/(x - 1)]$

1 mark
Invert and multiply.

$= 6 + (x + 5)(x - 1)/[(x + 5)(x - 2)]$

$= 6 + (x - 1)/(x - 2)$

$= [6(x - 2) + (x - 1)]/(x - 2)$

1 mark
Use a common denominator.

$= (7x - 13)/(x - 2)$

1 mark
a = 7, b = 13, c = 1, d = 2.

Domain: $x \neq 1, 2, -5$ for the original expression.

(Total for Question 13 is 4 marks)